P-TIWARI / Plug in, program, and build.
A.0 engineering prototype: see docs/Manufacturing_Readiness.md before fabrication.
Protected 2S battery; external balance charger. USB powers logic only.
Global nets join functional sheets. Four-layer board; 80 x 65 mm nominal.

GND
VBAT
LOGIC5
+3V3
+3V3_SENS
BAT5
VM
M_VINT

01 Controller 1

File: 01_Controller_1.kicad_sch

02 USB Programming 1

File: 02_USB_Programming_1.kicad_sch

03 Power Inputs 1

File: 03_Power_Inputs_1.kicad_sch

04 Logic Regulators 1

File: 04_Logic_Regulators_1.kicad_sch

05 Logic Regulators 2

File: 05_Logic_Regulators_2.kicad_sch

06 Servo Power 1

File: 06_Servo_Power_1.kicad_sch

07 Motor Driver 1

File: 07_Motor_Driver_1.kicad_sch

08 Motion Sensors 1

File: 08_Motion_Sensors_1.kicad_sch

09 Environmental Sensors 1

File: 09_Environmental_Sensors_1.kicad_sch

10 Distance Interfaces 1

File: 10_Distance_Interfaces_1.kicad_sch

11 Distance Interfaces 2

File: 11_Distance_Interfaces_2.kicad_sch

12 Encoder Interfaces 1

File: 12_Encoder_Interfaces_1.kicad_sch

13 Encoder Interfaces 2

File: 13_Encoder_Interfaces_2.kicad_sch

14 Servo PWM 1

File: 14_Servo_PWM_1.kicad_sch

15 GPIO Expansion 1

File: 15_GPIO_Expansion_1.kicad_sch

16 Analog Expansion 1

File: 16_Analog_Expansion_1.kicad_sch

17 User Interface 1

File: 17_User_Interface_1.kicad_sch

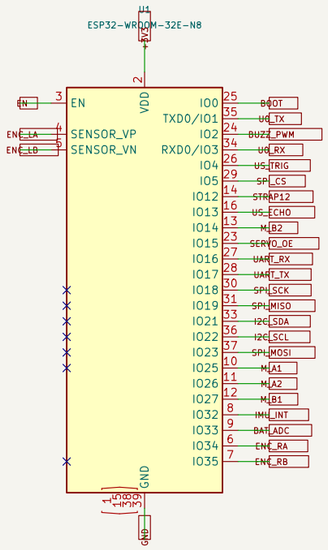
18 User Interface 2

File: 18_User_Interface_2.kicad_sch

Sheet: /
File: RoveCore_One.kicad_sch

Title: P-TIWARI | PTW-ESP32-ROB-A0

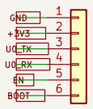
Size: A3	Date: 2026-11-08	**Rev: A.0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 1/19





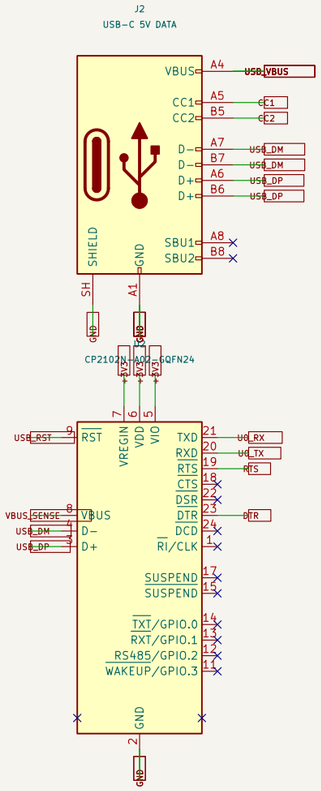


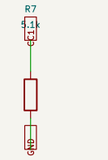


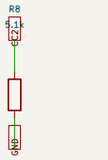


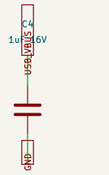
ESP32-WROOM-32E-N8: 8 MB flash; no PSRAM; USB bridge required.
GPIO34/35/36/39 input only. GPIO12 held LOW for 3.3V flash. Boot: hold BOOT, tap RESET.
All allocated GPIO are accessible at headers/test pads; shared signals are not free outputs.

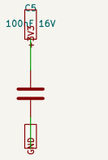
Sheet: /01 Controller 1/		
File: 01_Controller_1.kicad_sch		
Title: P-TIWARI | Controller		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 2/19

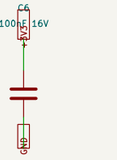


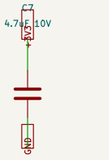




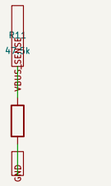


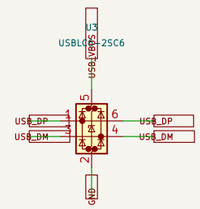


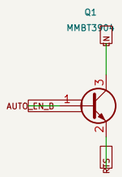


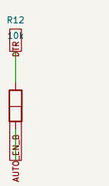


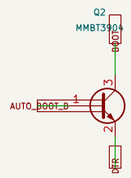


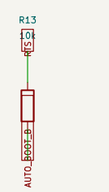










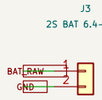


Self-powered CP2102N: VREGIN, VDD and VIO tied to 3V3 per Silicon Labs.
VBUS sensing uses 22.1k / 47.5k divider. Dual 5.1k CC pulldowns: 5V sink, no PD.
USB supplies LOGIC ONLY. Firmware must respect the host current entitlement.

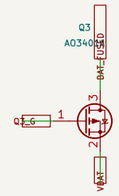
Sheet: /02 USB Programming 1/
File: 02_USB_Programming_1.kicad_sch

Title: P-TIWARI | USB Programming

Size: A3 | Date: 2026-11-08 | **Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6 | Id: 3/19



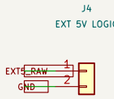


















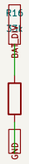












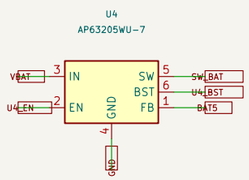


25 pack MUST include cell-level protection, overcurrent cutoff and balancing.
NO onboard charger: disconnect pack and charge with a proper 2S balance charger.
Schottky source OR prevents backfeeding USB/external/battery logic sources. Common ground.

Sheet: /03 Power Inputs 1/
File: 03_Power_Inputs_1.kicad_sch

Title: P-TIWARI | Power Inputs

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 4/19









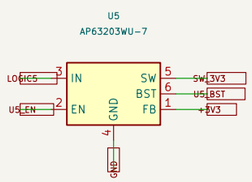


















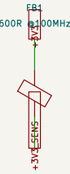














AP63205 battery buck 5V; AP63203 logic buck 3V3. Feedback on output side of inductors.
Battery EN divider 100k/27.4k: nominal rising UVLO about 5.6V. NOT cell protection.
3V3 target 1A total, sensor/expansion budget 150mA. Quiet sensor branch uses ferrite + local caps.

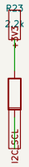
Sheet: /04 Logic Regulators 1/
File: 04_Logic_Regulators_1.kicad_sch

Title: P-TIWARI | Logic Regulators

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 5/19

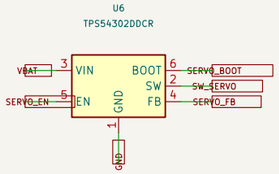






AP63205 battery buck 5V; AP63203 logic buck 3V3. Feedback on output side of inductors.
Battery EN divider 100k/27.4k: nominal rising UVLO about 5.6V, NOT cell protection.
3V3 target 1A total, sensor/expansion budget 150mA. Quiet sensor branch uses ferrite + local caps.

Sheet: /05 Logic Regulators 2/		
File: 05_Logic_Regulators_2.kicad_sch		
Title: P-TIWARI | Logic Regulators		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 6/19







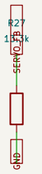








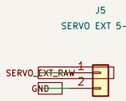


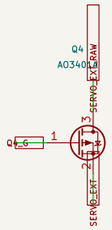




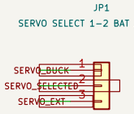
















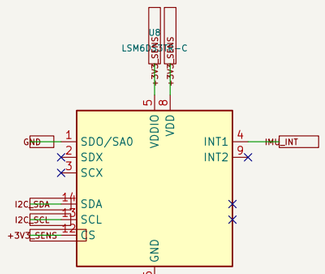
Servo source selector: fit ONE shunt only. Never bridge both supplies.
TPS54302 nominal Vout = 0.596*(1+100/13.3) = 5.077V. 10uH, 44uF ceramic, 75pF feedforward.
Servo recommendation: 2A total continuous, 3A brief peak ONLY after thermal validation. Four small servos, not four stall currents.

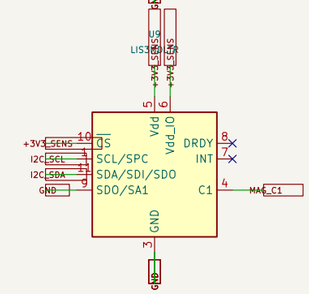
Sheet: /06 Servo Power 1/		
File: 06_Servo_Power_1.kicad_sch		
Title: P-TIWARI | Servo Power		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 7/19



DRV8833PWPR (PowerPAD), not uncooled PW. Rsense=0.25ohm => itrip typ 0.8A.
Recommend <=0.6A continuous each; 3–8.4V motor input. Limit is per bridge, not advertised 2A.
Regenerative braking raises VM; use a sink-capable battery supply. Validate VM stays <10.8V in operation, <12V absolute.
Fit 100nF directly at brushed motor terminals; keep twisted motor leads short.

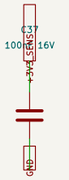
Sheet: /07 Motor Driver 1/		
File: 07_Motor_Driver_1.kicad_sch		
Title: P-TIWARI | Motor Driver		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 8/19





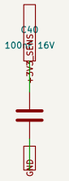












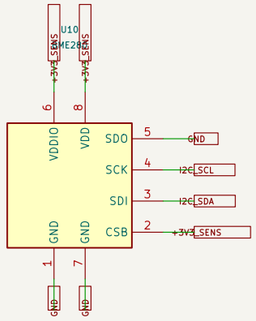


LSM6DS3TR-C uses installed LSM6DS3 symbol: identical 14-pin mapping, 3x2.5mm LGA footprint.
IMU I2C 0x6A (SA0=0), INT1 direct GPIO32. Magnetometer I2C 0x1C (SA1=0).
LIS3MDL reserved pins 2/12 grounded; C1 100nF. IMU SDx/SCx/INT2 intentionally unused.
Calibrate accelerometer, gyro bias and compass hard/soft iron in the assembled chassis.

Sheet: /08 Motion Sensors 1/
File: 08_Motion_Sensors_1.kicad_sch

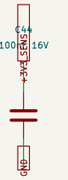
Title: P-TIWARI | Motion Sensors

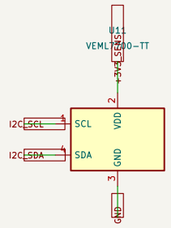
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 9/19









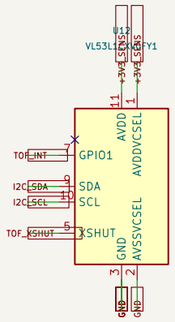


BME280 I2C 0x76: CSB=VDDIO, SDO=GND. VEML7700 fixed 0x10.
Keep optical/pressure openings clear. Do not wash MEMS/optical sensors without supplier approval.
BME temperature includes PCB self-heating; characterize board-on offsets under Wi-Fi/motor load.

Sheet: /09 Environmental Sensors 1/
File: 09_Environmental_Sensors_1.kicad_sch

Title: P-TIWARI | Environmental Sensors

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 10/19



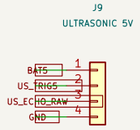


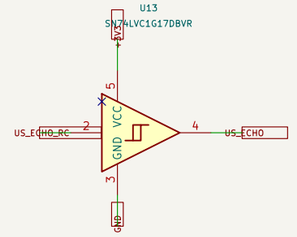
















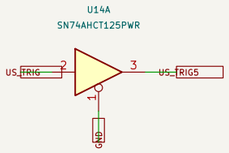


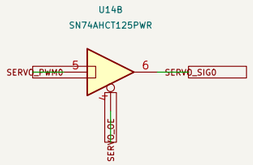


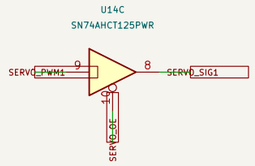


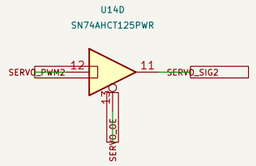
VL53L1X at 0x29: AVDD=3V3_SENS; program 2V8 IO mode with ST driver.
XSHUT open-drain control only. DNC pin8 intentionally open. Aperture faces upward: mount board to suit sensing direction.
Ultrasonic VCC is battery-derived BAT5: USB-only operation does not power 5V accessories.
Echo uses 5.5V-tolerant SN74LVC1G17 input; no direct 5V enters ESP32.

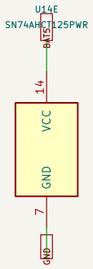
Sheet: /10 Distance Interfaces 1/		
File: 10_Distance_Interfaces_1.kicad_sch		
Title: P-TIWARI | Distance Interfaces		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 11/19









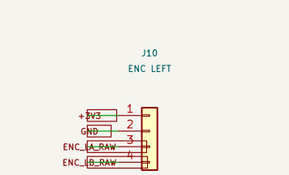


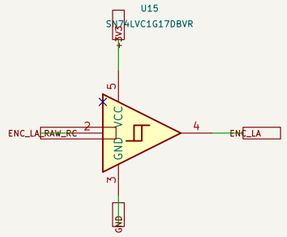
VL53L1X at 0x29: AVDD=3V3_SENS: program 2V8 IO mode with ST driver.
XSHUT open-drain control only. DNC pin8 intentionally open. Aperture faces upward: mount board to suit sensing direction.
Ultrasonic VCC is battery-derived BAT5: USB-only operation does not power 5V accessories.
Echo uses 5.5V-tolerant SN74LVC1G17 input; no direct 5V enters ESP32.

Sheet: /11 Distance Interfaces 2/
File: 11_Distance_Interfaces_2.kicad_sch

Title: P-TIWARI | Distance Interfaces

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 12/19



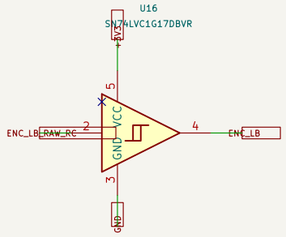










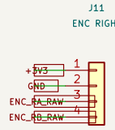


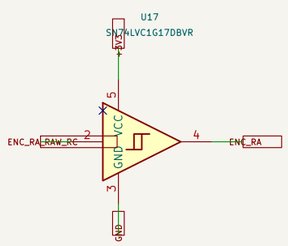










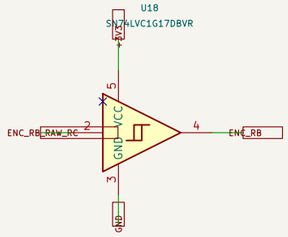
















Encoder connectors supply 3V3; push-pull or open-collector inputs accepted.
A/B inputs buffer-rated 0–5.5V, with 1k/100pF RC (~100ns) and 10k pullups.
GPIO36/39 left, GPIO34/35 right. Use ESP32 PCNT; no reliance on missing internal pullups.
External 5V encoder supply allowed only with shared ground and no tie to connector 3V3.

Sheet: /12 Encoder Interfaces 1/		
File: 12_Encoder_Interfaces_1.kicad_sch		
Title: P-TIWARI | Encoder Interfaces		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 13/19



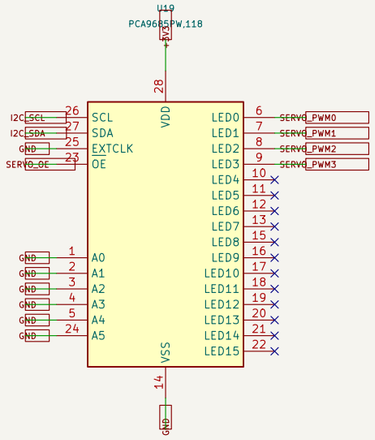


Encoder connectors supply 3V3; push-pull or open-collector inputs accepted.
A/B inputs buffer-rated 0-5.5V, with 1k/100pF RC (~100ns) and 10k pullups.
GPIO36/39 left, GPIO34/35 right. Use ESP32 PCNT; no reliance on missing internal pullups.
External 5V encoder supply allowed only with shared ground and no tie to connector 3V3.

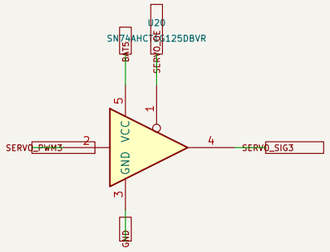
Sheet: /13 Encoder Interfaces 2/
File: 13_Encoder_Interfaces_2.kicad_sch

Title: P-TIWARI | Encoder Interfaces

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 14/19





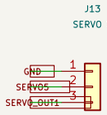




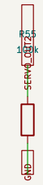


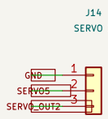






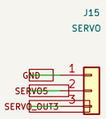






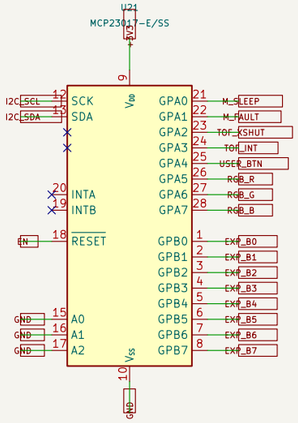




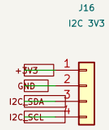


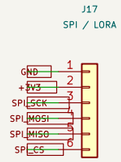
PCA9685 0x40: default all-call 0x70, disable if conflicting extension is installed.
GPIO15 high disables AHCT output buffers and PCA9685. Set low only after setting valid 50Hz pulses.
Each servo: GND / servo rail / 5V PWM. Servo supply and BAT5 logic are distinct.
Unused PWM channels intentionally NC; shared PWM frequency must be respected.

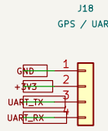
Sheet: /14 Servo PWM 1/	
File: 14_Servo_PWM_1.kicad_sch	
Title: P-TIWARI | Servo PWM	
Size: A3	Date: 2026-11-08
KiCad E.D.A. 10.0.6	**Rev: A.0 PROTOTYPE**
	Id: 15/19

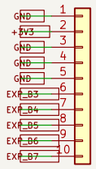




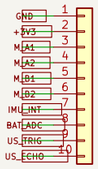










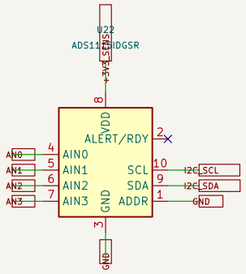


MCP23017 fixed 0x20. GPA0 sleep defaults LOW by external resistor; all pins reset as inputs.
GPA2 must be open-drain emulated: output LOW for shutdown, input for release; never drive HIGH.
GPB0..7 expansion are 3V3 GPIO; GPB7 also drives onboard load MOSFET.
GPIO headers expose SHARED nets: disable corresponding onboard function before reuse. GPIO12 not general expansion.

Sheet: /15 GPIO Expansion 1/
File: 15_GPIO_Expansion_1.kicad_sch

Title: P-TIWARI | GPIO Expansion

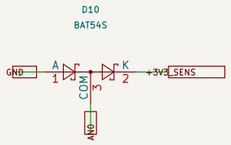
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 16/19

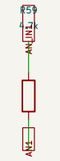




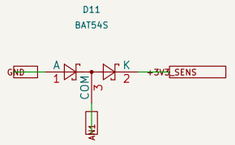






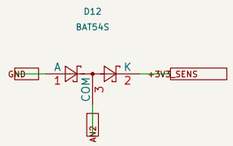






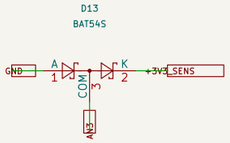


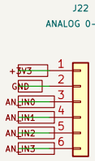






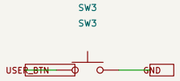






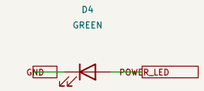
ADS1115 address 0x48, 860 samples/s aggregate maximum. Choose PGA +/−4.096V.
Input voltage range remains 0..3.3V; PGA range is NOT input tolerance. 4.7k / 10nF filtering with Schottky clamps.
Suitable for analog line sensor array. 4 channels sequentially sampled; high-speed line control may need an external ADC.

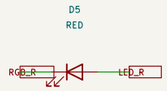
Sheet: /16 Analog Expansion 1/		
File: 16_Analog_Expansion_1.kicad_sch		
Title: P-TIWARI | Analog Expansion		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 17/19



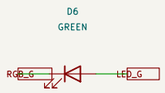


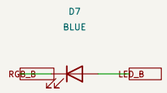


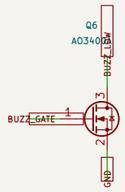


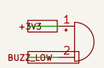




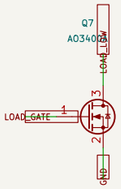


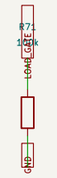








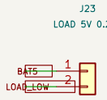






R/G/B discrete programmable LEDs on expander GPA5..7; use green for WI-FI status.
Piezo buzzer via GPIO2 MOSFET and discharge resistor. User button active LOW on GPA4.
GPB7 load output 5V/250mA max, diode-protected for small inductive loads. BAT5 accessory budget shared.
CAN: external 3V3 transceiver may reuse UART16/17 as TWAI RX/TX; GPS/UART then unavailable.

Sheet: /17 User Interface 1/		
File: 17_User_Interface_1.kicad_sch		
Title: P-TIWARI | User Interface		
Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 18/19



R/G/B discrete programmable LEDs on expander GPA5..7; use green for WI-FI status.
Piezo buzzer via GPIO2 MOSFET and discharge resistor. User button active LOW on GPA4.
GPB7 load output 5V/250mA max, diode-protected for small inductive loads. BAT5 accessory budget shared.
CAN: external 3V3 transceiver may reuse UART16/17 as TWAI RX/TX; GPS/UART then unavailable.

Sheet: /18 User Interface 2/
File: 18_User_Interface_2.kicad_sch

Title: P-TIWARI | User Interface

Size: A3	Date: 2026-11-08	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 19/19